Math 20

From UPD IM Module, 2018 — Module 0

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Contents

0 Notes **4**
- 0.1 Fractions . . . 4
- 0.2 Exponents . . . 5
- 0.3 Factoring . . . 6
- 0.4 Radicals . . . 7
- 0.5 Imaginary Numbers . . . 8
- 0.6 Complex Numbers . . . 9

1 **10**
- 1.1 . . . 10
- 1.2 . . . 11
- 1.3 . . . 12
- 1.4 . . . 13
- 1.5 . . . 14
- 1.6 . . . 15
- 1.7 . . . 16
- 1.8 . . . 17
- 1.9 . . . 18
- 1.10 . . . 19

2 **20**
- 2.1 . . . 20
- 2.2 . . . 21
- 2.3 . . . 22
- 2.4 . . . 23
- 2.5 . . . 24
- 2.6 . . . 25
- 2.7 . . . 26
- 2.8 . . . 27
- 2.9 . . . 28
- 2.10 . . . 29
- 2.11 . . . 30
- 2.12 . . . 31
- 2.13 . . . 32
- 2.14 . . . 33
- 2.15 . . . 34
- 2.16 . . . 35
- 2.17 . . . 36
- 2.18 . . . 37
- 2.19 . . . 38
- 2.20 . . . 39
- 2.21 . . . 40
- 2.22 . . . 41
- 2.23 . . . 42
- 2.24 . . . 43
- 2.25 . . . 44
- 2.26 . . . 45
- 2.27 . . . 46

3 **47**
- 3.1 . . . 47
- 3.2 . . . 48
- 3.3 . . . 49
- 3.4 . . . 50
- 3.5 . . . 51
- 3.6 . . . 52
- 3.7 . . . 53
- 3.8 . . . 54
- 3.9 . . . 55

3.10 56
3.11 57
3.12 58
3.13 59
3.14 60
3.15 61
3.16 62
3.17 63
3.18 64
3.19 65

4 **66**
4.1 66
4.2 67
4.3 68
4.4 69
4.5 70
4.6 71
4.7 72
4.8 73
4.9 74
4.10 75
4.11 76
4.12 77
4.13 78
4.14 79
4.15 80
4.16 81
4.17 82
4.18 83
4.19 84
4.20 85
4.21 86

5 **87**
5.1 87
5.2 88
5.3 89
5.4 90
5.5 91
5.6 92
5.7 93
5.8 94
5.9 95
5.10 96
5.11 97
5.12 98
5.13 99
5.14 100
5.15 101
5.16 102
5.17 103
5.18 104

0 Notes

0.1 Fractions

Note 0.1.1. Addition and Subtraction. Let L be the LCD of the fractions.

$$\frac{a}{b} \pm \frac{c}{d} = \frac{a\left(\frac{L}{b}\right) \pm c\left(\frac{L}{d}\right)}{L}$$

Note 0.1.2. Multiplication.

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

Note 0.1.3. Division.

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c} = \frac{ad}{bc}$$

If both fractions have the same denominator:

$$\frac{a}{b} \div \frac{c}{b} = \frac{a}{b} \cdot \frac{b}{c} = \frac{ab}{bc} = \frac{a}{c}$$

Note 0.1.4. Complex Fractions. Let L be the LCD of the fractions.

$$\frac{\frac{a}{b}}{\frac{c}{d}} = \frac{a\left(\frac{L}{b}\right)}{c\left(\frac{L}{d}\right)}$$

0.2 Exponents

Note 0.2.1. Negative Exponents.

$$a^{-n} = \frac{1}{a^n}$$

Note 0.2.2. Multiplication.

$$a^n a^m = a^{n+m}$$

Note 0.2.3. Division.

$$\frac{a^n}{a^m} = a^{n-m}$$

Note 0.2.4. Of Products.

$$(ab)^n = a^n b^n$$

Note 0.2.5. Of Quotients.

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

Note 0.2.6. Nesting.

$$(a^n)^m = a^{nm}$$

0.3 Factoring

Note 0.3.1. Common Monomial Factor.

$$ax \pm ay = a(x \pm y)$$

Note 0.3.2. Difference of Two Squares.

$$x^2 - y^2 = (x - y)(x + y)$$

Note 0.3.3. Perfect Square Trinomial.

$$x^2 \pm 2xy + y^2 = (x \pm y)^2$$

Note 0.3.4. Sum and Difference of Two Cubes.

$$x^3 \pm y^3 = (x \pm y)(x^2 \mp xy + y^2)$$

Note 0.3.5. Simple Trinomial.

$$x^2 + (a + b)x + ab = (x + a)(x + b)$$

Note 0.3.6. Complex Trinomial.

$$acx^2 + (ad + bc)x + bd = (ax + b)(cx + d)$$

0.4 Radicals

Note 0.4.1. Exponent Form.

$$\sqrt[n]{a^m} = a^{\frac{m}{n}}$$

Note 0.4.2. Of Products.

$$\sqrt[n]{ab} = \sqrt[n]{a}\,\sqrt[n]{b}$$

Note 0.4.3. Of Quotients.

$$\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

Note 0.4.4. Nesting.

$$\sqrt[n]{\sqrt[m]{a}} = \sqrt[nm]{a}$$

Note 0.4.5. Rationalization. A rational expression must not have a radical at its denominator. Multiply the numerator and denominator by an expression that would eliminate the radical; this may be a special product. A simple example:

$$\begin{aligned}
\sqrt[3]{\frac{10y^4}{3x^2}} &= \frac{\sqrt[3]{10y^4}}{\sqrt[3]{3x^2}} \\
&= \frac{\sqrt[3]{10y^4}}{\sqrt[3]{3x^2}} \cdot \frac{\sqrt[3]{9x}}{\sqrt[3]{3^2x}} \\
&= \frac{\sqrt[3]{90xy^4}}{\sqrt[3]{3^3x^3}} \\
&= \frac{\sqrt[3]{y^3}\,\sqrt[3]{90xy}}{3x} \\
&\boxed{= \frac{y\sqrt[3]{90xy}}{3x}}
\end{aligned}$$

A more complex example using note 0.3.4:

$$\begin{aligned}
\frac{1}{2-\sqrt[3]{4}} &= \frac{1}{2-\sqrt[3]{4}} \cdot \frac{4+2\sqrt[3]{4}+\sqrt[3]{16}}{4+2\sqrt[3]{4}+\sqrt[3]{4^2}} \\
&= \frac{4+2\sqrt[3]{4}+\sqrt[3]{8}\,\sqrt[3]{2}}{2^3-\sqrt[3]{4^3}} \\
&= \frac{4+2\sqrt[3]{4}+2\sqrt[3]{2}}{8-4} \\
&= \frac{4+2\sqrt[3]{4}+2\sqrt[3]{2}}{4} \\
&\boxed{= 1+\frac{\sqrt[3]{4}+\sqrt[3]{2}}{2}}
\end{aligned}$$

0.5 Imaginary Numbers

Note 0.5.1. Definition.

$$i = \sqrt{-1}$$

Note 0.5.2. Cyclicity. We have:

$$\begin{aligned} i^0 &= 1 \\ i^1 &= i \\ i^2 &= -1 \\ i^3 &= -i \end{aligned}$$

This is cyclic:

$$i^n = i^{n \bmod 4}$$

0.6 Complex Numbers

Note 0.6.1. Definition. A complex number in rectangular form is given by:

$$a + bi$$

Note 0.6.2. Addition.

$$(a + bi) + (c + di) = (a + c) + (b + d)i$$

Note 0.6.3. Multiplication.

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i$$

Note 0.6.4. Conjucate.

$$\overline{a + bi} = a - bi$$

Also,

$$(a + bi)(\overline{a + bi}) = a^2 + b^2$$

Note 0.6.5. Division.

$$\frac{a + bi}{c + di} = \frac{a + bi}{c + di} \cdot \frac{\overline{c + di}}{\overline{c + di}} = \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2} i$$

1

Simplify the following.

1.1

$$\frac{7^{-3}}{7^{-16}}$$

Note 0.2.3

$$= 7^{-3+16}$$

$$\boxed{= 7^{13}}$$

1.2

$$\frac{3^2 + 9^{-1} - 9^0}{2^4 - 3^{-1}}$$

$$= \left(9 + \frac{1}{9} - 1\right) \div \left(16 - \frac{1}{3}\right)$$

Notes 0.1.1 and 0.1.3

$$= \frac{81 + 1 - 9}{9} \cdot \frac{3}{48 - 1}$$
$$= \frac{73}{3} \cdot \frac{1}{47}$$

Note 0.1.2

$$\boxed{= \frac{73}{141}}$$

1.3

$$\frac{8x^6y^3z^4}{2x^5yz^2}$$

Note 0.2.3

$$= 4x^{6-5}y^{3-1}z^{4-2}$$

$$\boxed{= 4xy^2z^2}$$

1.4

$$\frac{3x^{21}y^{12}z^{19}}{(-x^{-9}y^{3}z^{-3})(-9x^{4}y^{6}z^{6})}$$

Note 0.2.2

$$= \frac{3x^{21}y^{12}z^{19}}{9x^{-9+4}y^{3+6}z^{-3+6}}$$
$$= \frac{3x^{21}y^{12}z^{19}}{9x^{-5}y^{9}z^{3}}$$

Note 0.2.3

$$= \frac{1}{3}x^{21+5}y^{12-9}z^{19-3}$$

$$\boxed{= \frac{1}{3}x^{26}y^{3}z^{16}}$$

1.5

$$\frac{2(x^3yz^2)^2}{(2x^2y^2z)^3}$$

Notes 0.2.4 and 0.2.6

$$= \frac{2x^6y^2z^4}{8x^6y^6z^3}$$

Note 0.2.3

$$= \frac{1}{4}x^{6-6}y^{2-6}z^{4-3}$$
$$= \frac{1}{4}y^{-4}z$$

Note 0.2.1

$$\boxed{= \frac{z}{4y^4}}$$

1.6

$$\frac{\left(2a^3b^{-3}c^{-2}\right)^3}{\left(a^3b^2c^{-1}\right)^{-2}}$$

Notes 0.2.4 and 0.2.6

$$= \frac{8a^9b^{-9}c^{-6}}{a^{-6}b^{-4}c^2}$$

Note 0.2.3

$$= 8a^{9+6}b^{-9+4}c^{-6-2}$$
$$= 8a^{15}b^{-5}c^{-8}$$

Note 0.2.1

$$\boxed{= \frac{8a^{15}}{b^5c^8}}$$

1.7

$$\left(\frac{2a^3b^{-3}c^{-1}}{3a^{-2}b^{-2}c^3}\right)^3$$

Note 0.2.3

$$= \left(\frac{2}{3}a^{3+2}b^{-3+2}c^{-1-3}\right)^3$$
$$= \left(\frac{2}{3}a^5b^{-1}c^{-4}\right)^3$$

Notes 0.2.4 to 0.2.6

$$= \frac{8}{27}a^{15}b^{-3}c^{-12}$$

Note 0.2.1

$$\boxed{= \frac{8a^{15}}{27b^3c^{12}}}$$

1.8

$$\left(\frac{a^2b^3c^{-1}}{ab^{-2}c^5}\right)^4$$

Note 0.2.3

$$= \left(a^{2-1}b^{3+2}c^{-1-5}\right)^4$$
$$= \left(ab^5c^{-6}\right)^4$$

Notes 0.2.4 and 0.2.6

$$= a^4b^{20}c^{-24}$$

Note 0.2.1

$$\boxed{= \frac{a^4b^{20}}{c^{24}}}$$

1.9

$$\left(\frac{4^0x^3y^{-2}z^{-4}}{3^2x^{-1}y^3z^{-5}}\right)^{-3}$$

$$= \left(\frac{x^3y^{-2}z^{-4}}{9x^{-1}y^3z^{-5}}\right)^{-3}$$

Note 0.2.3

$$= \left(\frac{1}{9}x^{3+1}y^{-2-3}z^{-4+5}\right)^{-3}$$

$$= \left(\frac{1}{9}x^4y^{-5}z\right)^{-3}$$

Notes 0.2.1 and 0.2.4 to 0.2.6

$$= 729x^{-12}y^{15}z^{-3}$$

Note 0.2.6

$$\boxed{= \frac{729y^{15}}{x^{12}z^3}}$$

1.10

$$\left(\frac{18^5 20^3 x^{-1} y^2 z^4}{15^9 x^{-3} y^{-2} z^{-1}}\right)^2$$

Notes 0.2.2 and 0.2.4

$$= \left(\frac{2^5 3^{10} 2^6 5^3 x^{-1} y^2 z^4}{3^9 5^9 x^{-3} y^{-2} z^{-1}}\right)^2$$

Note 0.2.2

$$= \left(\frac{2^{11} 3^{10} 5^3 x^{-1} y^2 z^4}{3^9 5^9 x^{-3} y^{-2} z^{-1}}\right)^2$$

Note 0.2.3

$$= (2^{11} 3^{10-9} 5^{3-9} x^{-1+3} y^{2+2} z^{4+1})^2$$
$$= (2^{11} 3^1 5^{-6} x^2 y^4 z^5)^2$$

Notes 0.2.4 and 0.2.6

$$= 2^{22} 3^2 5^{-12} x^4 y^8 z^{10}$$

Note 0.2.1

$$\boxed{= \frac{2^{22} 3^2 x^4 y^8 z^{10}}{5^{12}}}$$

2

Factor completely.

2.1

$$4x^3y^2 + 6x^2y^3$$

Note 0.3.1

$$\boxed{= 2x^2y^2(2x + 3y)}$$

2.2

$$a^2b^3c^4 - a^3b^4c^5 + 2a^2b^4c^4$$

Note 0.3.1

$$\boxed{= a^2b^3c^4(1 - abc + 2b)}$$

2.3

$$9a^2 - 1$$

Note 0.3.2

$$\boxed{= (3a - 1)(3a + 1)}$$

2.4

$$16x^2 - 25y^2$$

Note 0.3.2

$$\boxed{= (4x - 5y)(4x + 5y)}$$

2.5

$$25x^4y^8 - 64z^2$$

Note 0.3.2

$$\boxed{= (5x^2y^4 - 8z)(5x^2y^4 + 8z)}$$

2.6

$$16x^4 - 1$$

Note 0.3.2

$$= (4x^2 - 1)(4x^2 + 1)$$

Note 0.3.2

$$\boxed{= (2x - 1)(2x + 1)(4x^2 + 1)}$$

2.7

$$(s-2t)^2-4$$

Note 0.3.2

$$\boxed{=(s-2t-2)(s-2t+2)}$$

2.8

$$(x+3)^2 - (y+2)^2$$

Note 0.3.2

$$= (x+3-y-2)(x+3+y+2)$$

$$\boxed{= (x-y+1)(x+y+5)}$$

2.9

$$x^2(x^2-1)-9(x^2-1)$$

Note 0.3.1

$$=(x^2-9)(x^2-1)$$

Note 0.3.2

$$\boxed{=(x-3)(x+3)(x-1)(x+1)}$$

2.10

$$8x^9 + z^{12}$$

Note 0.3.4

$$\boxed{= (2x^3 + z^4)(4x^6 - 2x^3z^4 + z^8)}$$

2.11

$$8j^3 - 125k^6$$

Note 0.3.4

$$\boxed{= (2j - 5k^2)(4j^2 + 10jk^2 + 25k^4)}$$

2.12

$$(x-y)^3 - 1$$

Note 0.3.4

$$\boxed{= (x-y-1)((x-y)^2 + (x-y) + 1)}$$

2.13

$$x^2 - 2x - 3$$

Note 0.3.5

$$\boxed{= (x-3)(x+1)}$$

2.14

$$s^2 + 7s + 10$$

Note 0.3.5

$$\boxed{= (s+5)(s+2)}$$

2.15

$$48 - 13q - q^2$$

$$= -(q^2 + 13q - 48)$$

Note 0.3.5

$$\boxed{= -(q + 16)(q - 3)}$$

2.16

$$z^6 - 6z^3 - 72$$

Note 0.3.5

$$\boxed{= (z^3 - 12)(z^3 + 6)}$$

2.17

$$t^4 - 9st^2 - 10s^2$$

Note 0.3.5

$$\boxed{= (t^2 - 10s)(t^2 + s)}$$

2.18

$$a^6b^4 - a^3b^2c - 42c^2$$

Note 0.3.5

$$\boxed{= (a^3b^2 - 7c)(a^3b^2 + 6c)}$$

2.19

$$4x^2 - 4x + 1$$

Note 0.3.3

$$\boxed{= (2x-1)^2}$$

2.20

$$4n^2 - 12n + 9$$

Note 0.3.3

$$\boxed{= (2n-3)^2}$$

2.21

$$2r^2 - r - 3$$

Note 0.3.6

$$= 2r^2 - 3r + 2r - 3$$

$$\boxed{= (r+1)(2r-3)}$$

2.22

$$9a^2 + 15ab + 4b^2$$

Note 0.3.6

$$= 9a^2 + 12ab + 3ab + 4b^2$$

$$\boxed{= (3a + b)(3a + 4b)}$$

2.23

$$7s^6 - 9s^3 + 2$$

Note 0.3.6

$$\begin{aligned} &= 7s^6 - 7s^3 - 2s^3 + 2 \\ &= (7s^3 - 2)(s^3 - 1) \end{aligned}$$

Note 0.3.4

$$\boxed{= (7s^3 - 2)(s - 1)(s^2 + s + 1)}$$

2.24

$$a^2 + ab + 5a + 5b$$

Note 0.3.6

$$\boxed{= (a+5)(a+b)}$$

2.25

$$x^3 - x^2 - x + 1$$

Note 0.3.6

$$= (x^2 - 1)(x - 1)$$

Note 0.3.2

$$\boxed{= (x-1)^2(x+1)}$$

2.26

$$64x^4 + 1$$

$$= 64x^4 + 16x^2 + 1 - 16x^2$$

Note 0.3.3

$$= (8x^2 + 1)^2 - 16x^2$$

Note 0.3.2

$$\boxed{= (8x^2 - 4x + 1)(8x^2 + 4x + 1)}$$

2.27

$$4m^8 - 4m^4 + 9$$

$$= 4m^8 + 12m^4 + 9 - 16m^4$$

Note 0.3.3

$$= (2m^4 + 3)^2 - 16m^4$$

Note 0.3.2

$$\boxed{= (2m^4 - 4m + 3)(2m^4 + 4m + 3)}$$

3

Perform the indicated operations and simplify.

3.1

$$\frac{3a-3b}{c^2+3cd}\cdot\frac{2cd+6d^2}{ac-bc}$$

Note 0.3.1

$$=\frac{3(a-b)}{c(c+3d)}\cdot\frac{2d(c+3d)}{c(a-b)}$$

Note 0.1.2

$$\boxed{=\frac{6d}{c^2}}$$

3.2

$$\frac{x^3-8}{x^2-6x+8} \div \frac{3x^3+6x^2+12x}{x^2-8x+16}$$

Notes 0.1.3, 0.3.1 and 0.3.3 to 0.3.5

$$= \frac{(x-2)(x^2+2x+4)}{(x-4)(x-2)} \cdot \frac{(x-4)^2}{3x(x^2+2x+4)}$$

Note 0.1.2

$$\boxed{= \frac{x-4}{3x}}$$

3.3

$$\frac{x+1}{x} - \frac{y+1}{y}$$

Note 0.1.1

$$= \frac{xy + y - xy - x}{xy}$$

$$\boxed{= \frac{y-x}{xy}}$$

3.4

$$\frac{a}{bc} + \frac{b}{ac} - \frac{c}{ab}$$

Note 0.1.1

$$\boxed{= \frac{a^2 + b^2 - c^2}{abc}}$$

3.5

$$\frac{2}{x^2} + \frac{3}{xy} - \frac{1}{y^2}$$

Note 0.1.1

$$\boxed{= \frac{2y^2 + 3xy - x^2}{x^2y^2}}$$

3.6

$$\frac{1}{x} - \frac{2}{x+y}$$

Note 0.1.1

$$= \frac{x+y-2x}{x(x+y)}$$

$$\boxed{= \frac{y-x}{x(x+y)}}$$

3.7

$$\frac{2}{x-y} + \frac{3}{y-x}$$

$$= \frac{2}{x-y} - \frac{3}{x-y}$$

$$\boxed{= -\frac{1}{x-y}}$$

3.8

$$\frac{2}{(a+b)(a-b)} + \frac{1}{(a-b)^2}$$

Note 0.1.1

$$= \frac{2a - 2b + a + b}{(a+b)(a-b)^2}$$

$$\boxed{= \frac{3a - b}{(a+b)(a-b)^2}}$$

3.9

$$\frac{x^2}{x(x-1)} + \frac{1}{x} - \frac{1}{x-1}$$

Note 0.1.1

$$= \frac{x^2 + x - 1 - x}{x(x-1)}$$
$$= \frac{x^2 - 1}{x(x-1)}$$

Note 0.3.2

$$= \frac{(x-1)(x+1)}{x(x-1)}$$

$$\boxed{= \frac{x+1}{x}}$$

3.10

$$\frac{4}{4x^2-9} - \frac{3}{2x^2-x-3}$$

Notes 0.3.2 and 0.3.6

$$= \frac{4}{(2x-3)(2x+3)} - \frac{3}{2x^2-3x+2x-3}$$
$$= \frac{4}{(2x-3)(2x+3)} - \frac{3}{(x+1)(2x-3)}$$

Note 0.1.1

$$= \frac{4x+4-6x-9}{(2x-3)(2x+3)(x+1)}$$

$$\boxed{= -\frac{2x+5}{(2x-3)(2x+3)(x+1)}}$$

3.11

$$\frac{3}{2x-6} - \frac{9}{x^2-9}$$

Notes 0.3.1 and 0.3.2

$$= \frac{3}{2(x-3)} - \frac{9}{(x-3)(x+3)}$$

Note 0.1.1

$$= \frac{3x+9-18}{2(x-3)(x+3)}$$
$$= \frac{3x-9}{2(x-3)(x+3)}$$

Note 0.3.1

$$= \frac{3(x-3)}{2(x-3)(x+3)}$$

$$\boxed{= \frac{3}{2(x+3)}}$$

3.12

$$\frac{x - \frac{9}{x}}{1 - \frac{3}{x}}$$

Note 0.1.4

$$= \frac{x^2 - 9}{x - 3}$$

Note 0.3.2

$$= \frac{(x-3)(x+3)}{x-3}$$

$$\boxed{= x + 3}$$

3.13

$$\frac{2 - \frac{7}{x} - \frac{4}{x^2}}{3 - \frac{13}{x} + \frac{4}{x^2}}$$

Note 0.1.4

$$= \frac{2x^2 - 7x - 4}{3x^2 - 13x + 4}$$

Note 0.3.6

$$= \frac{2x^2 - 8x + x - 4}{3x^2 - 12x - x + 4}$$
$$= \frac{(2x+1)(x-4)}{(3x-1)(x-4)}$$

$$\boxed{= \frac{2x+1}{3x-1}}$$

3.14

$$\frac{w-2+\dfrac{w-2}{w+2}}{w-\dfrac{3w+12}{w+2}}$$

Note 0.1.4

$$= \frac{w^2+2w-2w-4+w-2}{w^2+2w-3w-12}$$
$$= \frac{w^2+w-6}{w^2-w-12}$$

Note 0.3.5

$$= \frac{(w+3)(w-2)}{(w-4)(w+3)}$$

$$\boxed{= \frac{w-2}{w-4}}$$

3.15

$$\frac{a^2 - b^2}{a^3 - b^3}$$

Notes 0.3.2 and 0.3.4

$$= \frac{(a-b)(a+b)}{(a-b)(a^2+ab+b^2)}$$

$$\boxed{= \frac{a+b}{a^2+ab+b^2}}$$

3.16

$$\frac{x^3 - x^2y + xy^2 - y^3}{x^6 + y^6}$$

Notes 0.3.4 and 0.3.6

$$= \frac{(x^2 + y^2)(x - y)}{(x^2 + y^2)(x^4 - x^2y^2 + y^4)}$$

$$\boxed{= \frac{x - y}{x^4 - x^2y^2 + y^4}}$$

3.17

$$\left(\frac{x}{x^2-1}-\frac{3}{x+1}\right)\div\frac{2x^2-x-3}{x^3-1}$$

Notes 0.1.3, 0.3.2, 0.3.4 and 0.3.6

$$\begin{aligned}
&= \left(\frac{x}{(x-1)(x+1)}-\frac{3}{x+1}\right)\cdot\frac{(x-1)(x^2+x+1)}{2x^2-3x+2x-3}\\
&= \left(\frac{x}{(x-1)(x+1)}-\frac{3}{x+1}\right)\cdot\frac{(x-1)(x^2+x+1)}{(x+1)(2x-3)}
\end{aligned}$$

Note 0.1.1

$$\begin{aligned}
&= \frac{x-3x+3}{(x-1)(x+1)}\cdot\frac{(x-1)(x^2+x+1)}{(x+1)(2x-3)}\\
&= -\frac{2x-3}{(x-1)(x+1)}\cdot\frac{(x-1)(x^2+x+1)}{(x+1)(2x-3)}
\end{aligned}$$

Note 0.1.2

$$\boxed{= -\frac{x^2+x+1}{(x+1)^2}}$$

3.18

$$\frac{\left(\frac{x}{x+y}+\frac{y}{x-y}\right)\left(\frac{x^2-xy}{x^4-y^4}\right)}{\frac{x}{x^2+2xy+y^2}}$$

Notes 0.1.1, 0.1.3 and 0.3.1 to 0.3.3

$$\begin{aligned}
&= \frac{x^2-xy+xy+y^2}{(x+y)(x-y)} \cdot \frac{x(x-y)}{(x^2-y^2)(x^2+y^2)} \cdot \frac{(x+y)^2}{x} \\
&= \frac{x^2+y^2}{(x+y)(x-y)} \cdot \frac{x(x-y)}{(x^2-y^2)(x^2+y^2)} \cdot \frac{(x+y)^2}{x}
\end{aligned}$$

Note 0.3.2

$$= \frac{x^2+y^2}{(x+y)(x-y)} \cdot \frac{x(x-y)}{(x-y)(x+y)(x^2+y^2)} \cdot \frac{(x+y)^2}{x}$$

Note 0.1.2

$$\boxed{= \frac{1}{x-y}}$$

3.19

$$\frac{\dfrac{3}{p+q}+\dfrac{1}{p-2q}}{1+\dfrac{p-q}{p-2q}}$$

Notes 0.1.1 and 0.1.3

$$\begin{aligned}
&= \frac{3p-6q+p+q}{(p+q)(p-2q)} \cdot \frac{p-2q}{p-2q+p-q} \\
&= \frac{4p-5q}{(p+q)(p-2q)} \cdot \frac{p-2q}{2p-3q}
\end{aligned}$$

Note 0.1.2

$$\boxed{= \frac{4p-5q}{(p+q)(2p-3q)}}$$

4

Perform the indicated operations and simplify.

4.1

$$\frac{b^{\frac{2}{5}}b^{\frac{1}{3}}}{b^{\frac{2}{15}}}$$

Notes 0.1.1 and 0.2.2

$$= \frac{b^{\frac{6+5}{15}}}{b^{\frac{2}{15}}}$$
$$= \frac{b^{\frac{11}{15}}}{b^{\frac{2}{15}}}$$

Notes 0.1.1 and 0.2.3

$$= b^{\frac{11-2}{15}}$$
$$= b^{\frac{9}{15}}$$

$$\boxed{= b^{\frac{3}{5}}}$$

4.2

$$\frac{24c^{-\frac{1}{2}}d^{\frac{2}{3}}}{18c^{-\frac{1}{7}}d^{-\frac{3}{5}}}$$

Notes 0.1.1 and 0.2.3

$$= \frac{4}{3}c^{\frac{-7+2}{14}}d^{\frac{10+9}{15}}$$
$$= \frac{4}{3}c^{-\frac{5}{14}}d^{\frac{19}{15}}$$

Note 0.2.1

$$= \frac{4d^{\frac{19}{15}}}{3c^{\frac{5}{14}}}$$

4.3

$$\frac{\left(2^{\frac{2}{3}}x^{\frac{1}{2}}y^{\frac{1}{6}}z\right)^6}{4x^{\frac{2}{3}}yz^{\frac{9}{2}}}$$

Notes 0.1.2, 0.2.4 and 0.2.6

$$= \frac{16x^3yz^6}{4x^{\frac{2}{3}}yz^{\frac{9}{2}}}$$

Notes 0.1.1 and 0.2.3

$$= 4x^{\frac{9-2}{3}}z^{\frac{12-9}{2}}$$

$$\boxed{= 4x^{\frac{7}{3}}z^{\frac{3}{2}}}$$

4.4

$$(u^{\frac{1}{3}} + (uv)^{\frac{1}{6}} + v^{\frac{1}{3}})(u^{\frac{1}{6}} - v^{\frac{1}{6}})$$

Note 0.3.4

$$\boxed{= (u^{\frac{1}{2}} - v^{\frac{1}{2}})}$$

4.5

$$\left(\frac{r^{\frac{4}{3}}s^{\frac{1}{2}}t^{-2}}{rs^{\frac{2}{7}}t^{\frac{3}{5}}}\right)^{-10}$$

Notes 0.1.1 and 0.2.3

$$= (r^{\frac{4-3}{3}}s^{\frac{7-4}{14}}t^{\frac{-10-3}{5}})^{-10}$$
$$= (r^{\frac{1}{3}}s^{\frac{3}{14}}t^{-\frac{13}{5}})^{-10}$$

Notes 0.2.4 and 0.2.6

$$= r^{-\frac{10}{3}}s^{-\frac{30}{14}}t^{26}$$
$$= r^{-\frac{10}{3}}s^{-\frac{15}{7}}t^{26}$$

Note 0.2.1

$$\boxed{= \frac{t^{26}}{r^{\frac{10}{3}}s^{\frac{15}{7}}}}$$

4.6

$$\sqrt[3]{-8^4}$$

Note 0.4.2

$$= \sqrt[3]{-8}\sqrt[3]{8^3}$$

$$\boxed{= -16}$$

4.7

$$\sqrt{384x^3y^6}$$

Note 0.4.2

$$= \sqrt{64x^2y^6}\sqrt{6x}$$

$$\boxed{= 8xy^3\sqrt{6x}}$$

4.8

$$\sqrt[4]{9x^8}$$

Note 0.4.2

$$= \sqrt[4]{x^8}\sqrt[4]{3^2}$$

$$\boxed{= x^2\sqrt{3}}$$

4.9

$$\sqrt{\sqrt[3]{4a^4b^4}}$$

Note 0.4.4

$$= \sqrt[6]{2^2a^4b^4}$$

$$\boxed{= \sqrt[3]{2a^2b^2}}$$

4.10

$$\sqrt{\frac{25c^5}{12a^3b}}$$

Note 0.4.3

$$= \frac{\sqrt{25c^5}}{\sqrt{12a^3b}}$$

Note 0.4.2

$$= \frac{\sqrt{25c^4}\sqrt{c}}{\sqrt{4a^2}\sqrt{3ab}}$$
$$= \frac{5c^2\sqrt{c}}{2a\sqrt{3ab}}$$

Note 0.4.5

$$= \frac{5c^2\sqrt{c}}{2a\sqrt{3ab}} \cdot \frac{\sqrt{3ab}}{\sqrt{3ab}}$$

Note 0.1.2

$$\boxed{= \frac{5c^2\sqrt{3abc}}{6a^2b}}$$

4.11

$$\frac{2x^2y}{\sqrt[3]{12x^2y^4}}$$

Note 0.4.2

$$= \frac{2x^2y}{\sqrt[3]{y^3}\,\sqrt[3]{12x^2y}}$$
$$= \frac{2x^2}{\sqrt[3]{12x^2y}}$$

Note 0.4.5

$$= \frac{2x^2}{\sqrt[3]{12x^2y}} \cdot \frac{\sqrt[3]{144xy^2}}{\sqrt[3]{12^2xy^2}}$$

Note 0.1.2

$$= \frac{2x^2\,\sqrt[3]{144xy^2}}{12xy}$$

Note 0.4.2

$$= \frac{2x^2\,\sqrt[3]{8}\,\sqrt[3]{18xy^2}}{12xy}$$
$$= \frac{4x^2\,\sqrt[3]{18xy^2}}{12xy}$$

$$\boxed{= \frac{x\,\sqrt[3]{18xy^2}}{3y}}$$

4.12

$$\sqrt{2}\sqrt[3]{2}$$

Note 0.4.1

$$= 2^{\frac{1}{2}}2^{\frac{1}{3}}$$

Notes 0.1.1 and 0.2.2

$$= 2^{\frac{3+2}{6}}$$

$$\boxed{= 2^{\frac{5}{6}}}$$

4.13

$$\frac{x^2-1}{\sqrt{x+1}}$$

Note 0.3.2

$$= \frac{(x-1)(x+1)}{\sqrt{x+1}}$$

Note 0.4.5

$$= \frac{(x-1)(x+1)}{\sqrt{x+1}} \cdot \frac{\sqrt{x+1}}{\sqrt{x+1}}$$

Note 0.1.2

$$= \frac{(x-1)(x+1)\sqrt{x+1}}{x+1}$$

$$\boxed{= (x-1)\sqrt{x+1}}$$

4.14

$$\frac{\sqrt{a^2-b^2}}{\sqrt{a^2+b^2}}$$

Note 0.4.5

$$= \frac{\sqrt{a^2-b^2}}{\sqrt{a^2+b^2}} \cdot \frac{\sqrt{a^2+b^2}}{\sqrt{a^2+b^2}}$$

Notes 0.1.2, 0.3.2 and 0.4.2

$$\boxed{= \frac{\sqrt{a^4-b^4}}{a^2+b^2}}$$

4.15

$$\sqrt[4]{36} - \sqrt{54} + \sqrt{96}$$

Note 0.4.2

$$\begin{aligned} &= \sqrt[4]{6^2} - \sqrt{9}\sqrt{6} + \sqrt{16}\sqrt{6} \\ &= \sqrt{6} - 3\sqrt{6} + 4\sqrt{6} \end{aligned}$$

$$\boxed{= 2\sqrt{6}}$$

4.16

$$\frac{2-\sqrt{3}}{3-\sqrt{2}}$$

Notes 0.3.2 and 0.4.5

$$= \frac{2-\sqrt{3}}{3-\sqrt{2}} \cdot \frac{3+\sqrt{2}}{3+\sqrt{2}}$$

Note 0.1.2

$$= \frac{(2-\sqrt{3})(3+\sqrt{2})}{9-2}$$

$$\boxed{= \frac{6+2\sqrt{2}-3\sqrt{3}-\sqrt{6}}{7}}$$

4.17

$$\frac{\sqrt{6}-\sqrt{5}}{\sqrt{5}+\sqrt{6}}$$

Notes 0.3.2 and 0.4.5

$$= \frac{\sqrt{6}-\sqrt{5}}{\sqrt{5}+\sqrt{6}} \cdot \frac{\sqrt{5}-\sqrt{6}}{\sqrt{5}-\sqrt{6}}$$

Note 0.1.2

$$= \frac{(\sqrt{6}-\sqrt{5})(\sqrt{5}-\sqrt{6})}{5-6}$$
$$= (\sqrt{6}-\sqrt{5})^2$$

Note 0.3.3

$$\boxed{= 11 - 2\sqrt{30}}$$

4.18

$$\frac{x^2 - 2x + 1}{\sqrt{x} + 1}$$

Note 0.3.3

$$= \frac{(x-1)^2}{\sqrt{x} + 1}$$

Notes 0.3.2 and 0.4.5

$$= \frac{(x-1)^2}{\sqrt{x} + 1} \cdot \frac{\sqrt{x} - 1}{\sqrt{x} - 1}$$

Note 0.1.2

$$= \frac{(x-1)^2(\sqrt{x} - 1)}{x - 1}$$

$$\boxed{= (x-1)(\sqrt{x} - 1)}$$

4.19

$$\frac{6}{\sqrt{3-\sqrt{3}}}$$

Note 0.4.5

$$= \frac{6}{\sqrt{3-\sqrt{3}}} \cdot \frac{\sqrt{3-\sqrt{3}}}{\sqrt{3-\sqrt{3}}}$$

Note 0.1.2

$$= \frac{6\sqrt{3-\sqrt{3}}}{3-\sqrt{3}}$$

Notes 0.3.2 and 0.4.5

$$= \frac{6\sqrt{3-\sqrt{3}}}{3-\sqrt{3}} \cdot \frac{3+\sqrt{3}}{3+\sqrt{3}}$$

Note 0.1.2

$$= \frac{6(3+\sqrt{3})\sqrt{3-\sqrt{3}}}{9-3}$$

$$\boxed{= (3+\sqrt{3})\sqrt{3-\sqrt{3}}}$$

4.20

$$\frac{1}{\sqrt[3]{4}+\sqrt[3]{-27}}$$

$$= \frac{1}{\sqrt[3]{4}-3}$$

Notes 0.3.4 and 0.4.5

$$= \frac{1}{\sqrt[3]{4}-3} \cdot \frac{\sqrt[3]{16}+3\sqrt[3]{4}+9}{\sqrt[3]{16}+3\sqrt[3]{4}+9}$$

Note 0.1.2

$$= \frac{\sqrt[3]{16}+3\sqrt[3]{4}+9}{4-27}$$

$$\boxed{= -\frac{\sqrt[3]{16}+3\sqrt[3]{4}+9}{23}}$$

4.21

$$\frac{2\sqrt{5}}{\sqrt{8}} + \frac{9}{\sqrt[3]{16}}$$

Note 0.4.3

$$= \frac{2\sqrt{5}}{\sqrt{4}\sqrt{2}} + \frac{9}{\sqrt[3]{8}\sqrt[3]{2}}$$
$$= \frac{2\sqrt{5}}{2\sqrt{2}} + \frac{9}{2\sqrt[3]{2}}$$

Note 0.4.5

$$= \frac{2\sqrt{5}}{2\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} + \frac{9}{2\sqrt[3]{2}} \cdot \frac{\sqrt[3]{4}}{\sqrt[3]{2^2}}$$

Note 0.1.2

$$= \frac{2\sqrt{10}}{4} + \frac{9\sqrt[3]{4}}{4}$$

$$\boxed{= \frac{2\sqrt{10} + 9\sqrt[3]{4}}{4}}$$

5

Perform the indicated operations and simplify.

5.1

$$\overline{1-i}$$

$$=$$

5.2

$$\overline{5i+2}$$

$$=$$

5.3

$$(\overline{5-i}) - (\overline{5+i})$$

$$=$$

5.4

$$\sqrt{-1} - \sqrt{3} + \sqrt{2} - \sqrt{-4}$$

$$=$$

5.5

$$4 - 2\sqrt{-45} - (\overline{3 - \sqrt{-20}})$$

$$=$$

5.6

$$\sqrt{-12}\sqrt{-75}$$

$$=$$

5.7

$$3i(i^2 - i^3 + 5i^5 - i^{-2})$$

$$=$$

5.8

$$(3-5i)(7+4i)$$

$$=$$

5.9

$$(1+2i)(3-4i)$$

$$=$$

5.10

$$(i-1)(\overline{5+3i})$$

$$=$$

5.11

$$(2i-3)^2$$

$$=$$

5.12

$$\frac{3+4i}{2-i}$$

$$=$$

5.13

$$\frac{\sqrt{-9}+\sqrt{8}}{\sqrt{-1}}$$

$$=$$

5.14

$$\frac{3i-2}{3i+2}$$

$$=$$

5.15

$$\frac{(1-i)(3i+1)}{2i-1}$$

$$=$$

5.16

$$\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}i\right)\left(\frac{\sqrt{2}}{2}-\frac{\sqrt{2}}{2}i\right)$$

$$=$$

5.17

$$\frac{7+i-4(3-i)}{6-5i^3}$$

$$=$$

5.18

$$\frac{2-2(\overline{i+1})}{2-\sqrt{-4}}$$

$$=$$